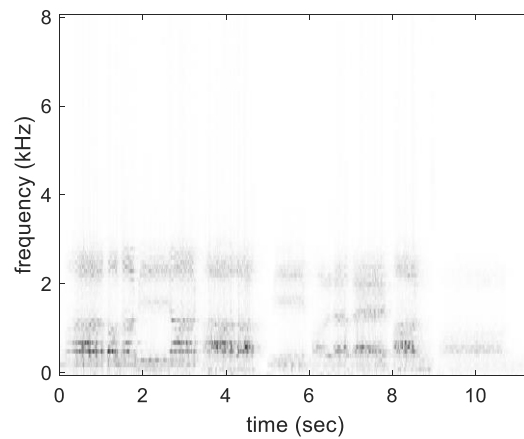
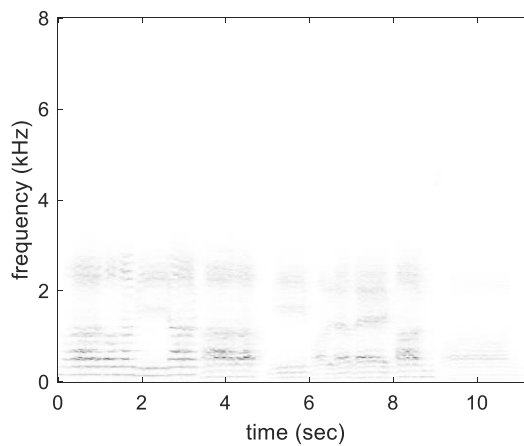


1.

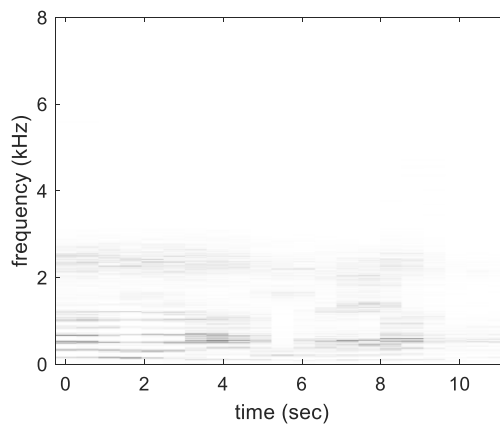
For the male voice



(frame length = 0.01 sec)



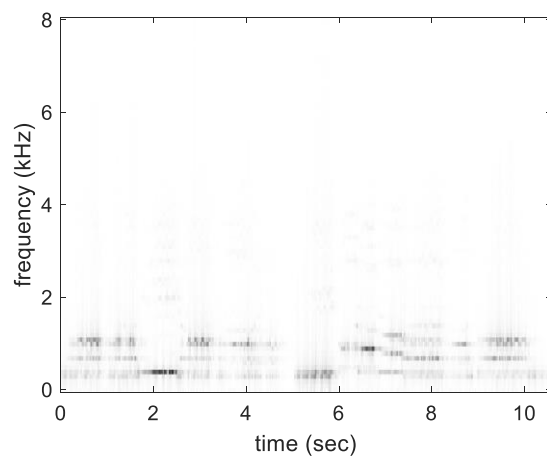
(frame length = 0.1 sec)



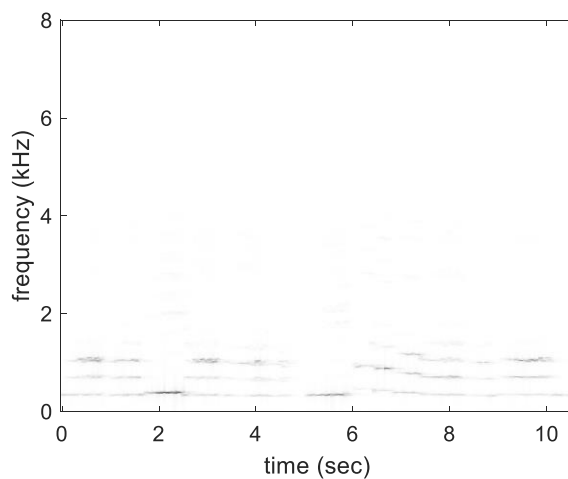
(frame length = 1 sec)

Comparing the above three spectrograms, we can observe that the frame length affects frequency and time resolution. The longer frame length gives rise to some obvious side lobe effects on y-direction due to its high frequency resolution, although we can roughly recognize where the harmonic frequency is. The shorter one results in ambiguous spectrum such that harmonic frequency is difficult to be perceived. Hence, 0.1 second frame length is worthy to deal with the male voice.

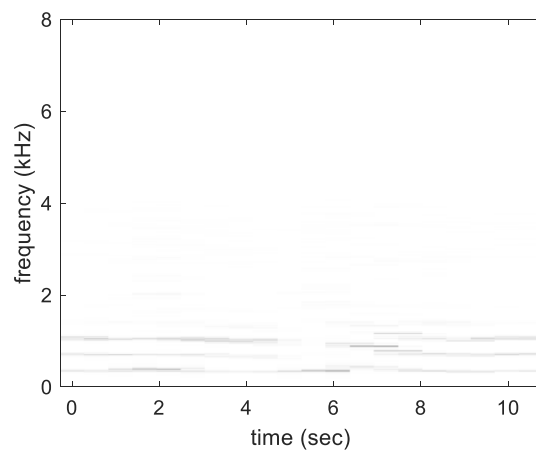
For the female voice



(frame length = 0.01 sec)



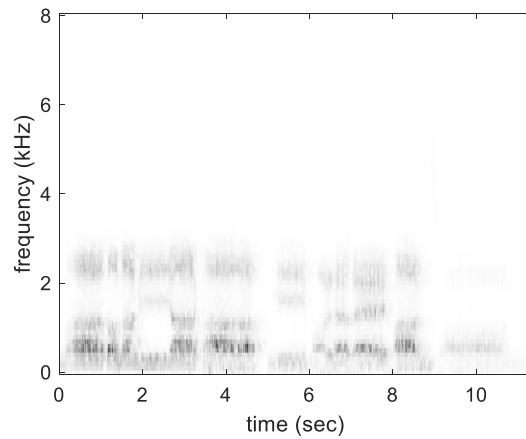
(frame length = 0.1 sec)



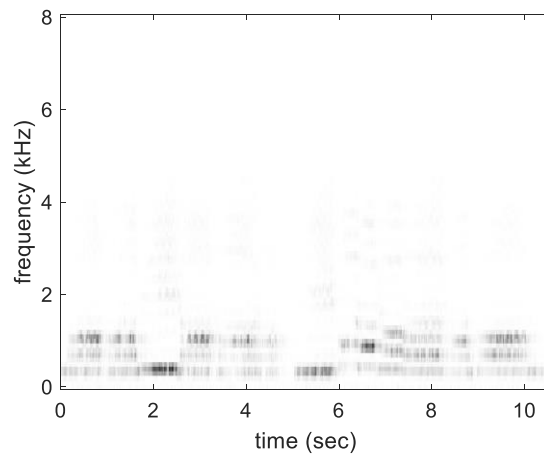
(frame length = 1 sec)

We encounter the same situation, so about 0.1 second frame length is appropriate for the female voice. In my perspective, the results generated from 0.05 second to 1 second frame length are able to see the harmonics clearly. Their acceptability depends on the person.

- 2.
- In mathematically, using Hann window instead of rectangular window will get slighter side lobes and a wider main lobe.



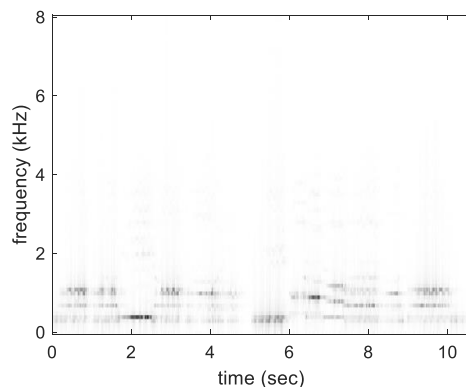
(frame length = 0.01 sec male voice)



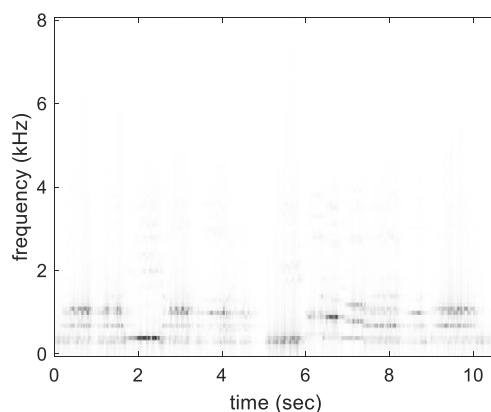
(frame length = 0.01 sec female voice)

Compare the above two figures to the previous results. The harmonic frequencies generated by Hann window has a wider spread than one from rectangular window as our expected. Hence, if the width of main lobe is too long, harmonics may overlap such that you cannot see separation frequency components. And it affects you to choose frame length.

3.



(female voice with frame length = 0.01 sec and sw.emphasis = 1)

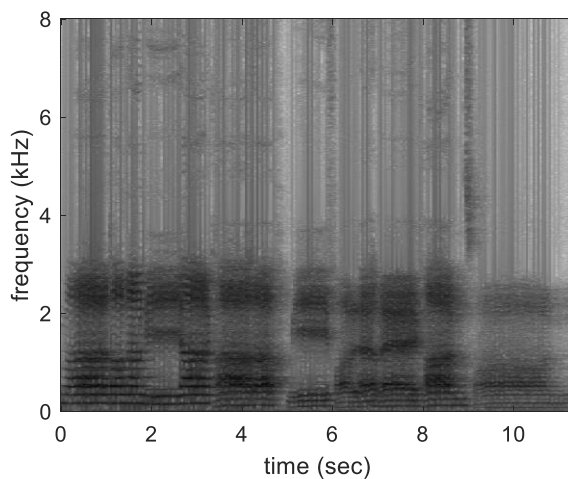


(female voice with frame length = 0.01 sec and sw.emphasis = 0)

To be honest, I cannot even notice any difference between the two figures whereas the comment says it boots up high frequency components. It's also useless that I try to change the frame length and voice source.

4.

All above figures are displayed in `abs(S)` rather than log conversion. If we turn to do log conversion, the spectrogram consists of dark splines. In my opinion, the display way, `abs(S)`, is intuitively better to find the harmonics because we need high contrast.



5.

Once we have to detect where the extreme points locate, because they can help us to find peak value. One of the properties of an extreme point is where its derivative equals zero. In discrete time, that is equivalently to calculate its difference and find where the sign changes.

```
% find peak
Ymag = Ymag+offset;
Y_diff = Ymag(1:end-1) - Ymag(2:end); % calculate difference
```

The bandwidth of human voice is given. In order to reduce computation complexity, only low frequency components below 1000Hz are interested.

```
changesign = (Y_diff(1:end-1,:).*Y_diff(2:end,:))<0; % verify where the sign changes
changesign = [false;changesign;false]; % padding zero to compensate the lack of indice
cutoff_freq = 1000; % to reduce computation then convert it to index
changesign = changesign(1:ceil(cutoff_freq/df));
ff_extreme = ff(changesign); % find freq of turning points
val_extreme = Ymag(changesign); % find value of turning points
```

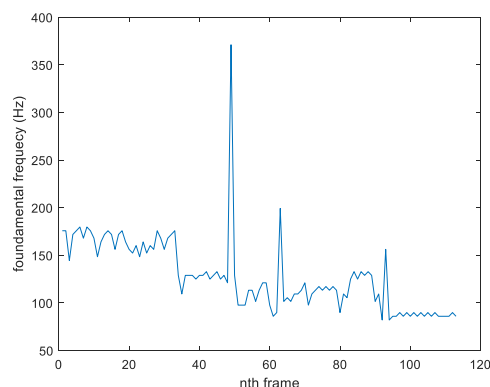
After extreme point detection, we are required to classify the particularly large ones. They may be the fundamental frequency or harmonics. Standard deviation and average are helpful to find these outliers, so we can set n standard deviation as a threshold to do it, where n is a constant determined by trial and error.

```
sigma = std(val_extreme);
mu = mean(val_extreme);
threshold = mu + 0.5*sigma;
```

The first outlier can be viewed as the fundamental frequency if it's over 80Hz for male or 150Hz for female. Otherwise, the second outlier is regarded as the fundamental frequency and so on.

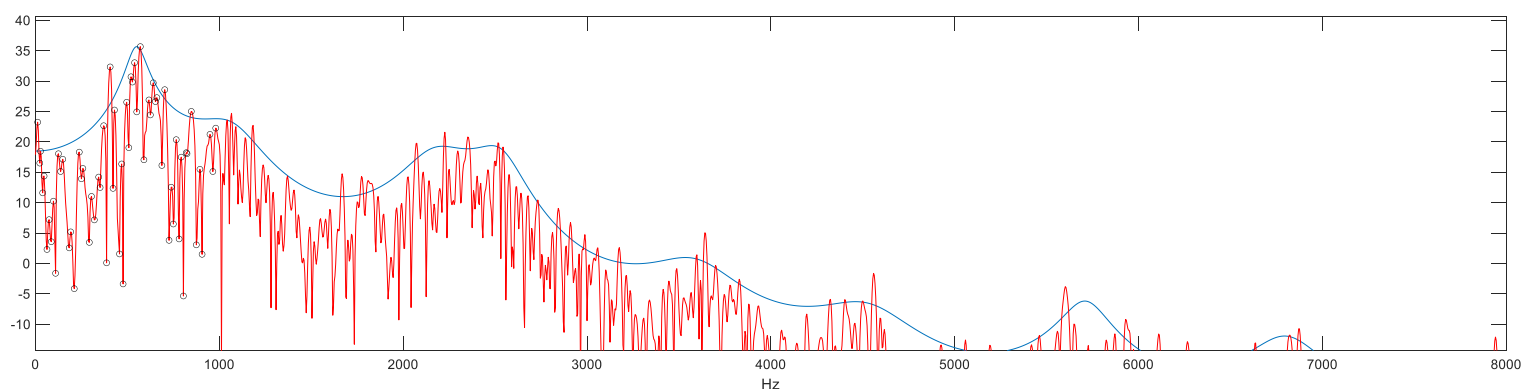
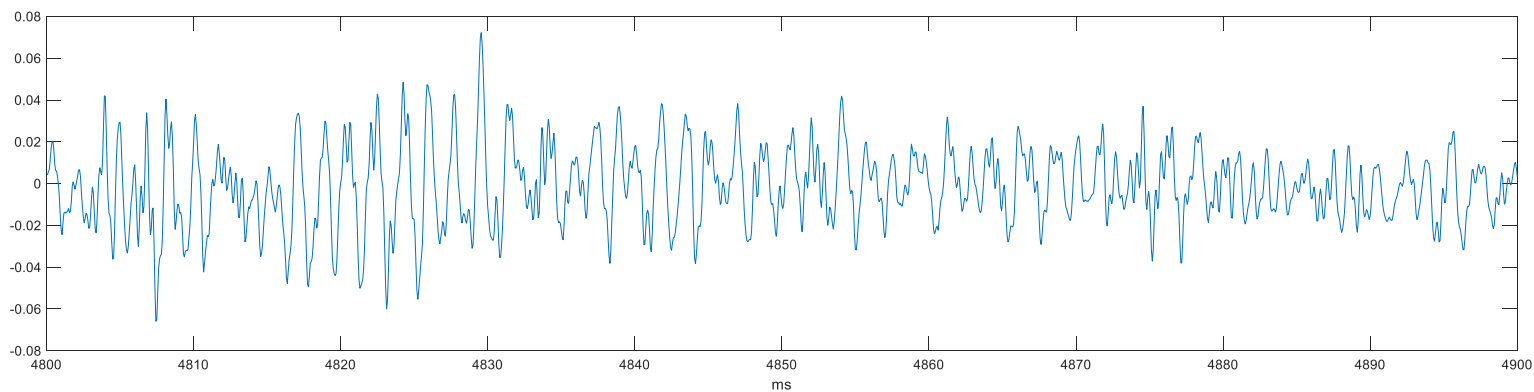
```
ff_extreme = ff_extreme(val_extreme > threshold);
jj=1;
fundamental(kk) = ff_extreme(jj);
% for male
while fundamental(kk) < 80
    jj = jj + 1;
    fundamental(kk) = ff_extreme(jj);
end
% for female
while fundamental(kk) < 150
    jj = jj + 1;
    fundamental(kk) = ff_extreme(jj);
end
```

The following picture shows the pitch contour of male voice.



(pitch contour of male voice with frame length = 0.1)

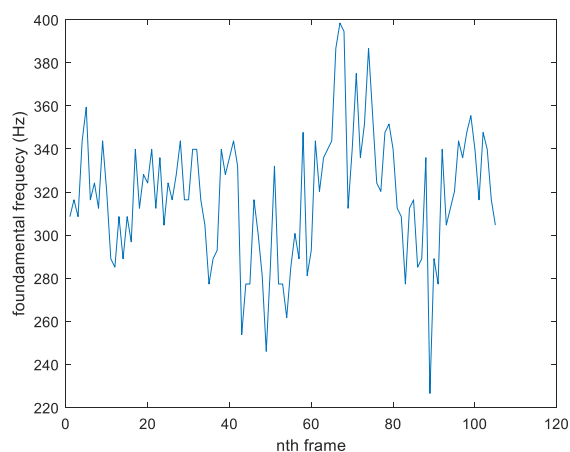
We can see there is an abnormal fundamental frequency at the 49th frame, checking what the spectrum looks like.



(spectrum of the 49th frame)

In my opinion, this spectrum is difficult to directly find the fundamental frequency not to mention our algorithm. It should be forgiven. Except for the 49th frame, rest of frames have an elegant performance, because they are in the range of 80 to 320 Hz.

The following picture shows the pitch contour of female voice.



(pitch contour of female voice with frame length = 0.1)

It seems perfect without any abnormal element. The pitch contour of neither female nor male voice is under A4 or 440Hz.

6.

The matter is the threshold determined by your experience, so we have to try the best circumstance iteratively. Using the first frame or randomly selected to manually check whether the fundamental frequency is over the threshold. The pitch contour figure helps us to check which frame is not applicable. Finally, we modify the threshold and repeat the process until the performance is satisfied. If we alternate using log conversion referred to question 4, it will be useless. After log conversion, the small values are magnified, and large values are shrunk. It leads us to set the constant n difficultly. That says this algorithm lacks robust. However, the process is particularly successful and every thing is exactly what I expect.